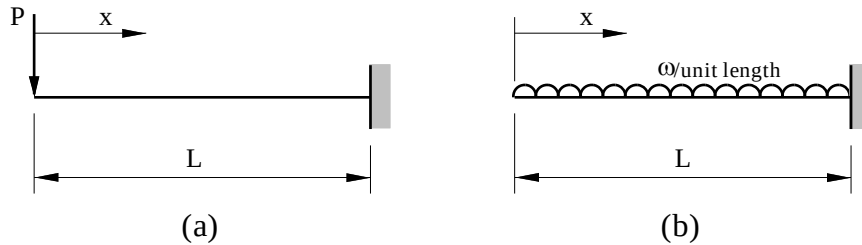


Mechanics 1

Tutorial Sheet - 5

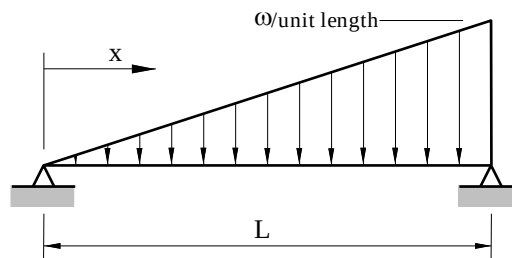
1. The uniform cantilevers shown in the diagram are each of length L and have flexural rigidity EI . Note that at the fixed end both the deflection and the slope are zero : $v = dv/dx = 0$.



In each case, derive expressions for the deflection and the slope at the free end.

Ans: (a) $v_{x=0} = \frac{PL^3}{3EI}$, $\left(\frac{dv}{dx}\right)_{x=0} = -\frac{PL^2}{2EI}$ (b) $v_{x=0} = \frac{\omega L^4}{8EI}$, $\left(\frac{dv}{dx}\right)_{x=0} = -\frac{\omega L^3}{6EI}$

2. The simply-supported beam shown in the diagram, span L and flexural rigidity EI , carries a distributed load which varies linearly from zero at one end of the beam to ω /unit length at the other.



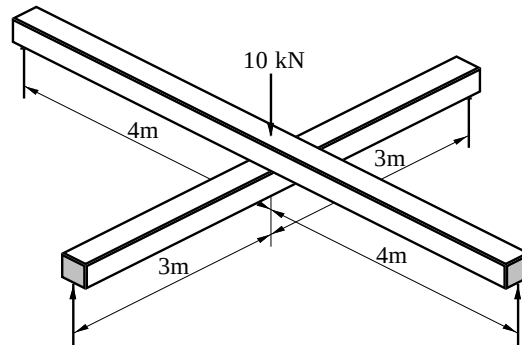
Show that the bending moment and deflection of the beam are given by:

$$M = \frac{\omega L x}{6} - \frac{\omega x^3}{6L} \quad ; \quad v = \frac{\omega L^4}{EI} \left\{ \frac{x^5}{120L^5} - \frac{x^3}{36L^3} + \frac{7x}{360L} \right\}$$

and hence find expressions for the position and magnitude of the largest deflection.

Ans: $x = 0.519L$, $v_{\max} = 0.00652 \frac{\omega L^4}{EI}$

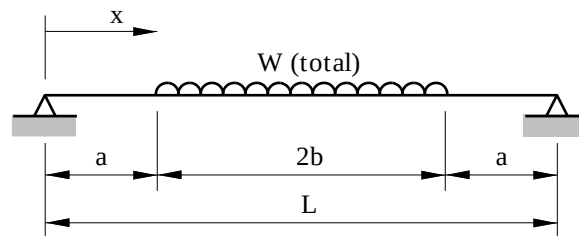
3. The diagram shows two beams, simply-supported at their ends, with the same flexural rigidity EI , which cross at right angles at their mid-points - the upper beam just resting upon the lower. The upper beam carries a concentrated load of 10 kN at mid-span. Note that the deflections of the beams at the point of contact must be identical.



What is the reaction between the beams at the point of contact and what is the greatest bending moment in each of the beams?

Ans: 7.03 kN; (Upper) 5.94 kNm; (Lower) 10.55 kNm

4. A uniform simply supported beam, span L and flexural rigidity EI , carries a total load W , uniformly distributed over the central portion of the span ($w = W/2b$), as shown in the diagram.

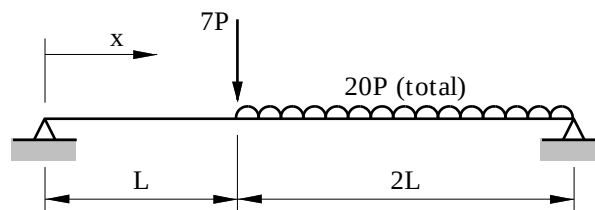


Form an expression for the bending moment in the *left-hand* half of the beam ($x \leq L/2$) and show that the maximum deflection (at mid-span) is given by:

$$\delta = \frac{W}{48EI} (L^3 - 2b^2L + b^3)$$

Confirm that this corresponds to the known results for the limiting cases: $b = 0$ and $b = L/2$.

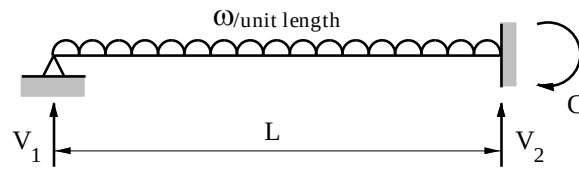
5. A uniform simply-supported beam, of span $3L$ and flexural rigidity EI , carries a concentrated load $7P$ and a uniformly distributed load $20P$ (total), as shown in the diagram.



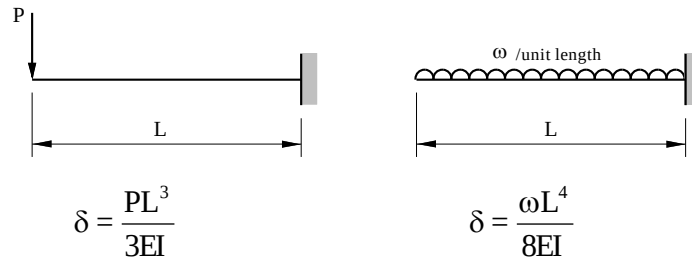
Form an expression for the bending moment as a single function of x , and show that the maximum deflection occurs at mid-span ($x = 3L/2$). What is the magnitude of the maximum deflection?

Ans: $\delta = \frac{723PL^3}{64EI}$

6. The diagram shows a 'propped cantilever', of length L and uniform flexural rigidity EI , fully fixed at one end and simply-supported at the other. A uniformly distributed load, ω , acts on the full length.



Recall that the deflections at the free end of an *unpropped* cantilever carrying a concentrated load and a uniformly distributed load are given by :



What are the external reactions at the supports of the propped cantilever? Draw the Shear Force and Bending Moment Diagrams and find the greatest bending moment in the beam.

Ans: $V_1 = \frac{3}{8}\omega L$; $V_2 = \frac{5}{8}\omega L$; $C = \frac{1}{8}\omega L^2$; $M_{\max} = +\frac{9}{128}\omega L^2, -\frac{1}{8}\omega L^2$